
Self-trigger, with full-stream fidelity

Find pulses in the FPGA.
Match them in the DAQ.

DAPHNE / PDS

15 September 2026

DUNE
DEEP UNDERGROUND
NEUTRINO EXPERIMENT

 **Fermilab**

The second pulse continues in the next window

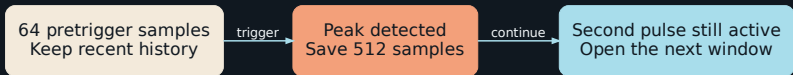
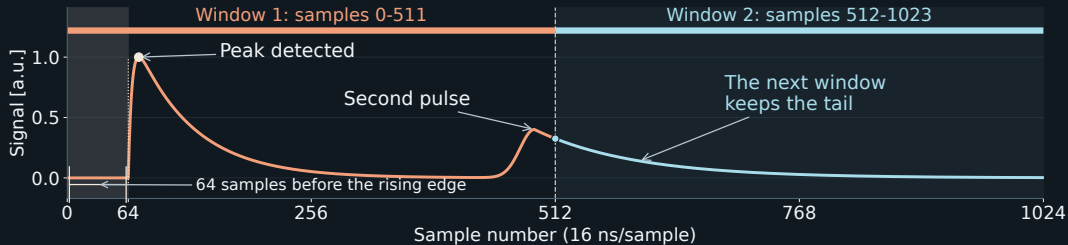


Diagram. Illustration: the boundary splits the record, while continuation keeps the second pulse's tail.

The milestone is agreement with full stream

THE GOAL

Keep the same signal.

Make it easier to trigger on.

Match signal timing and charge, while giving the DAQ peak descriptors for online coincidences.

Full stream leaves pulse finding to the DAQ



Diagram. Raw waveforms still need online pulse finding before the DAQ can compare photon times.

Raw streaming is a computing problem

32 CHANNELS AT 62.5 MS/s

Two billion samples every second.

The DAQ must find pulses in that stream.

FPGA peak descriptors let the online trigger start from pulse information.

Per-board sample workload.

Local FD documents separate photon readout and trigger processing; they do not give a measured PDS pulse-finding limit.

We have an storage limit with computing group.

Self-trigger moves that work into the FPGA

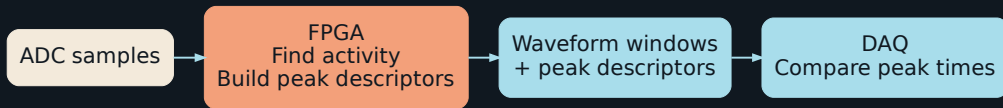


Diagram. Peak descriptors travel with the waveform.
The DAQ can use their times without repeating pulse finding on every sample.

Coincidences can form local PDS trigger activity

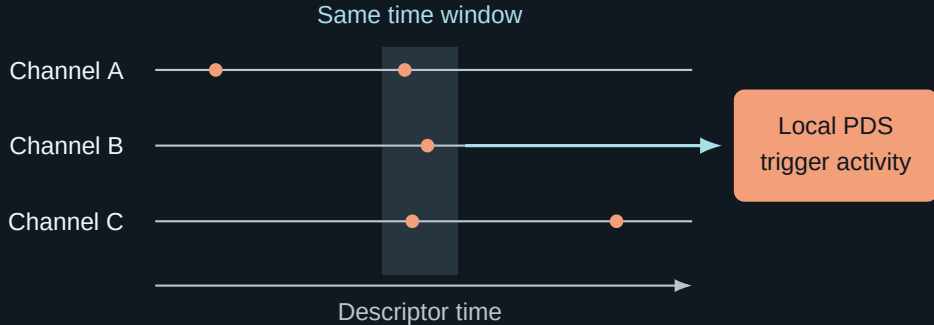


Diagram. Illustration. Compare times across channels on a shared clock.
Tune the coincidence window and multiplicity using signal and background studies.

Four channels share one builder

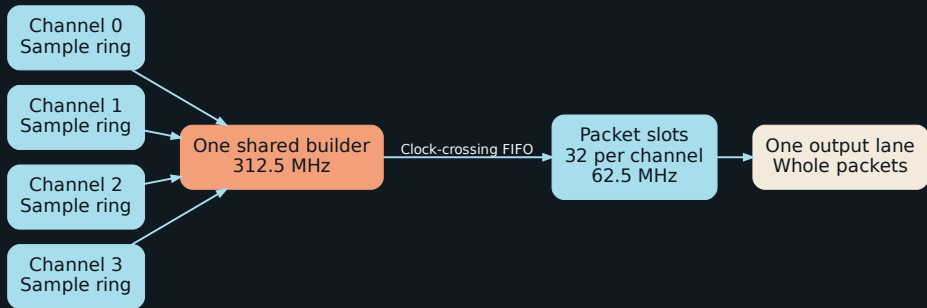


Diagram. Each builder packs waveform records and calculates peak descriptors. Eight builders serve 32 channels.

Replay the same signals through all three

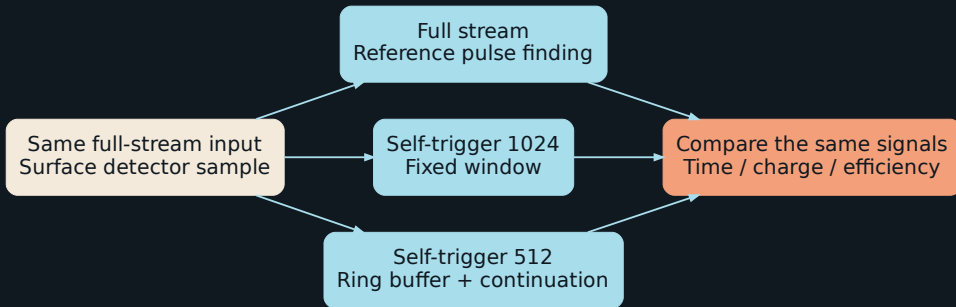


Diagram. Use the same surface full-stream input, thresholds, and output limits.
Compare recovered pulses, peak times, and charge. Matched three-way results are still needed.

The peak descriptors match the input waveform

FRESH RTL REPLAY

379,656 descriptor words checked.

An independent calculation checks each saved record's peak summaries against the known input.

Diagram. Five deterministic cases. This checks descriptor correctness; it does not yet test local coincidence activity.

Saving samples is not the same as finding pulses

Readout	Sample retention	Pulse finding
Self-trigger 1024	Limited by busy windows	FPGA descriptors
Self-trigger 512 + ring	Buffers reduce losses	FPGA descriptors
Full stream	Every sample*	DAQ must find the peaks

Full stream is the sample reference.

Self-trigger moves pulse finding into the FPGA.

Diagram. * Assuming lossless transport and storage. Saving all samples does not guarantee finding every pulse online.

More peak slots cost readout capacity

Samples / peak slots	Bytes / frame	Missing descriptors	Dense lane use
Current 512 / 5	960	50.3%	95.31%
Compact 512 / 5	952	50.3%	94.53%
256 / 2	480	61.3%	96.88%
256 / 3	488	44.9%	98.44%
256 / 5	504	18.9%	101.56%

$$U_{\text{lane}} = \frac{4(B/8 + 2)}{N} \times 100\% \quad U_{256/5} = \frac{4(63 + 2)}{256} \times 100\% = 101.56\%$$

B : frame bytes; N : samples. Four channels share one internal lane.

One 64-bit word per cycle, plus two cycles per frame; continuous capture.

Above 100%: sustained backlog. Intermittent capture uses less capacity.

Diagram. Descriptor loss: one synthetic case, 40 detected photons/channel/burst, $1.6 \mu\text{s}$ slow light, extrapolated +5 V afterpulsing. Simplified peak-count model; waveform charge retained.

LArSoft supplies the detector-side load

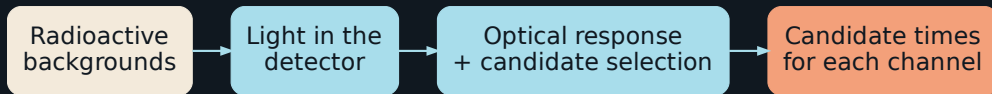


Diagram. Use physical activity to test pulse finding and coincidences. The grouped RTL replay currently uses synthetic waveforms.

The VD cathode sees the most activity

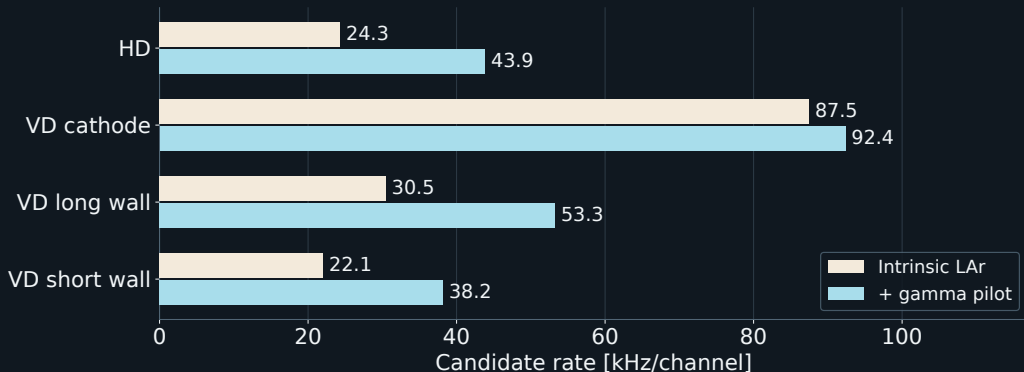


Diagram. Mean candidate rates above 0.7 PE. Intrinsic LAr: eight shards. Gamma addition: one window per source family, still a pilot.

Show the same signals in both readouts

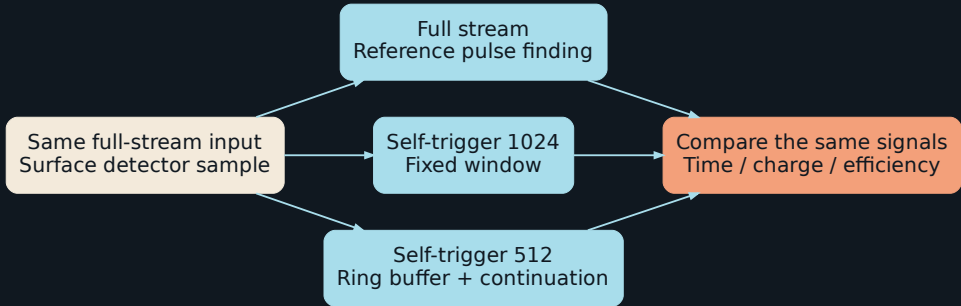


Diagram. The milestone test: matched waveforms and backgrounds, then matched signal timing, charge, and local trigger activity.

BACKUP

Why this helps

Dead time and low-energy physics without a beam gate.

Buffers buy time

SHORT STALLS

Keep collecting while the output waits.

The ring holds recent samples. Reserved packet slots hold complete records. Long stalls eventually fill both.

Continuation keeps the tail

LONG PULSES

Add another 512 samples when needed.

Short pulses stay short. Sustained activity can span adjacent records. Check recovered charge to see whether the tail survives.

We cannot wait for the beam

LOW-ENERGY PHYSICS

Keep the light from weak signals.

Preserving low-energy light signals can support background rejection and searches for solar and supernova neutrinos.

Diagram. Study question: can better background tagging let us use more of the detector?

Source: [DUNE Phase II: low-energy physics opportunities](#)

Start with the surface full-stream data

A PESSIMISTIC STRESS TEST

We already have a demanding sample in xc_sim.

Replay it through self-trigger. Compare the saved waveform, peak times, and charge against the raw data.

Diagram. Surface-detector activity tests robustness under a harsh load. It does not set the underground far-detector background rate.

Each pulse gets a start, height, area, and duration

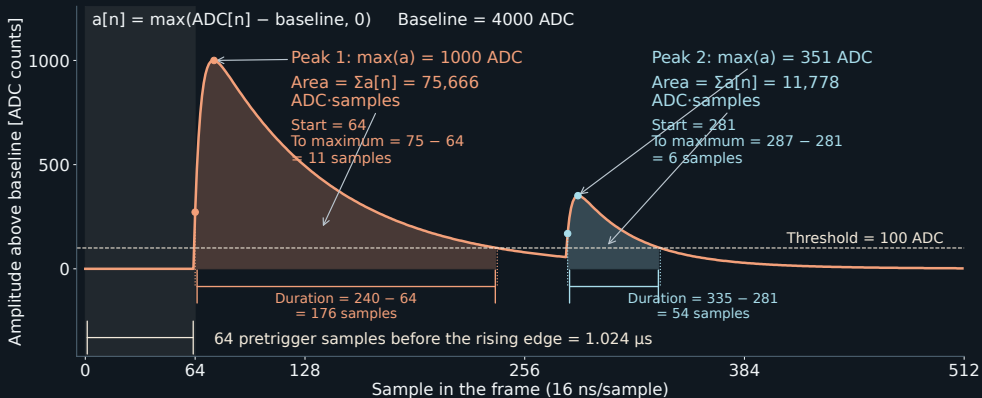


Diagram. Sum amplitudes above threshold; do not subtract the threshold. Both slots valid; format byte 0xF1. Illustrative waveform.

Five peak descriptors now cover half the waveform

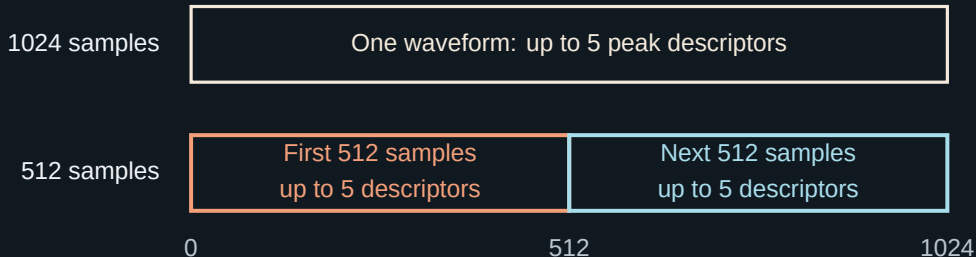


Diagram. The same five-peak format gives finer descriptor coverage: up to ten peak summaries over 1024 samples when both fragments are retained.

Every common packet matches

Fresh five-mode RTL replay; firmware 7a25777e

Mode	Grouped packets	Native packets	Common / identical	Max latency [μ s]
0	3,392	3,392	3,392	17.984
1	3,372	3,372	3,372	17.952
2	2,224	2,200	2,200	560.432
3	3,672	3,446	944	48.272
4	3,296	3,272	3,272	17.984

13,180 common packets are byte-identical; 16,198,656 ADC samples checked across both implementations.

Mode 4 includes deliberate reset and disable losses. Latency runs from the first sample to the final output word.

Count saved samples, not just packets

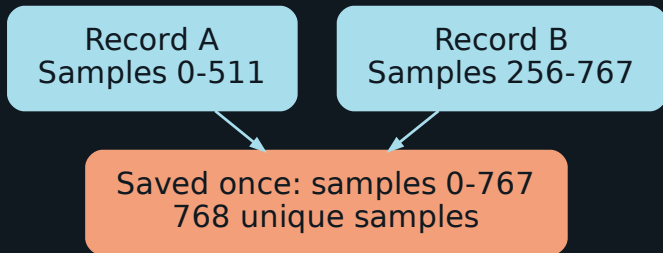


Diagram. Merge overlapping windows before measuring loss. Each nonzero input sample counts once; quiet samples are outside the denominator.

Stop the output and watch what happens

ONE STRESS TEST

Pause all lanes for 544 microseconds.

Let the buffers fill. Restart readout. Check what was kept, what was lost, and whether it recovers.

Diagram. Other cases check simultaneous pulses, short stalls, overlapping triggers, and reset.

Eight lanes feed four links

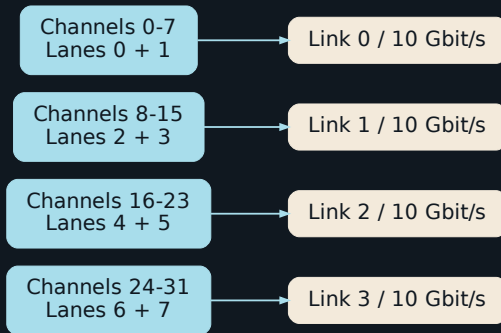


Diagram. Grouped32 mapping. The replay checks the lanes; it does not send data through physical Ethernet.

Full stream also pairs two streams per link

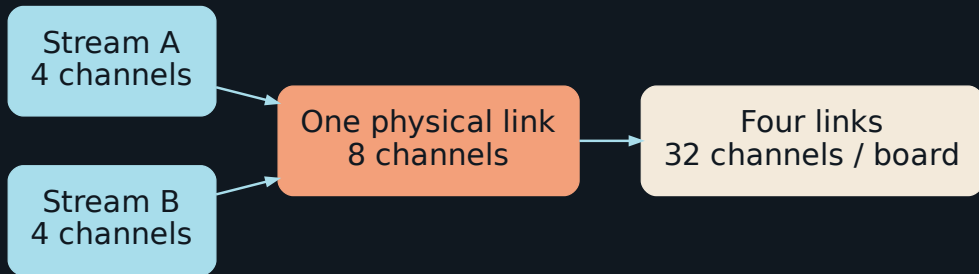


Diagram. Checked full-stream source: four channels per logical stream, eight per physical link, 32 per board. Installed image and channel selection were not checked.

Expected traffic from the modeled activity only

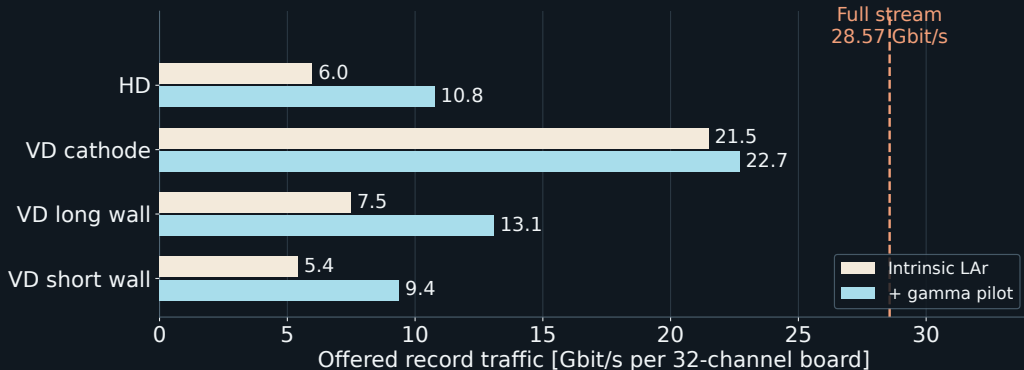


Diagram. Other background sources are not included. 32 channels; one self-trigger record per candidate, before losses or continuation. Both estimates exclude network overhead.

Full stream is constant; self-trigger follows activity

Same 32-channel board	Full stream	Self-trigger 512 + ring
What is sent?	Every sample	Selected windows + peaks
What sets the rate?	ADC sample clock	Saved records per second
HD example below	28.57 Gbit/s	10.78 Gbit/s

HD intrinsic + gamma pilot: 43,855 candidates/s/channel.

$$32 \times 43,855 \times 960 \text{ bytes} \times 8 = 10.78 \text{ Gbit/s.}$$

Diagram. Estimate: one 512-sample record per candidate; no continuation, merging, or losses modeled. Other backgrounds are excluded. Both rates include record headers, not network overhead.

Preprocessing matters even when traffic stays high

CONTINUOUS ACTIVITY

The descriptors still save pulse-finding work.

Continuous self-trigger records reach 30 Gbit/s for 32 channels. Bandwidth savings can disappear while the preprocessing benefit remains.

The LArSoft numbers are a dated snapshot

Software	dunesw v10_22_00d01; larsoft v10_22_00
Detector models	FD horizontal drift (HD), vertical drift (VD)
Candidate selection	Above 0.7 photoelectrons (PE); no 10 MeV cut
Intrinsic baseline	Atmospheric argon mixture; eight shards
Gamma addition	One physical window per source family
Snapshot	9 September 2026; reused here, not rerun

Shared transport, a complete background inventory, and calibrated noise are still needed for a detector trigger prediction.

Keep the result tied to its inputs

Firmware: 7a25777e, grouped32 branch.

Run: `scripts/run_grouped32_rtl_study.py -run`.

Evidence: this deck's README, JSON reports, CSV tables, and figure scripts.

Behavioral memory and clock-crossing models; no physical Ethernet or analog capture test. Source: [Firmware architecture and verification](#)